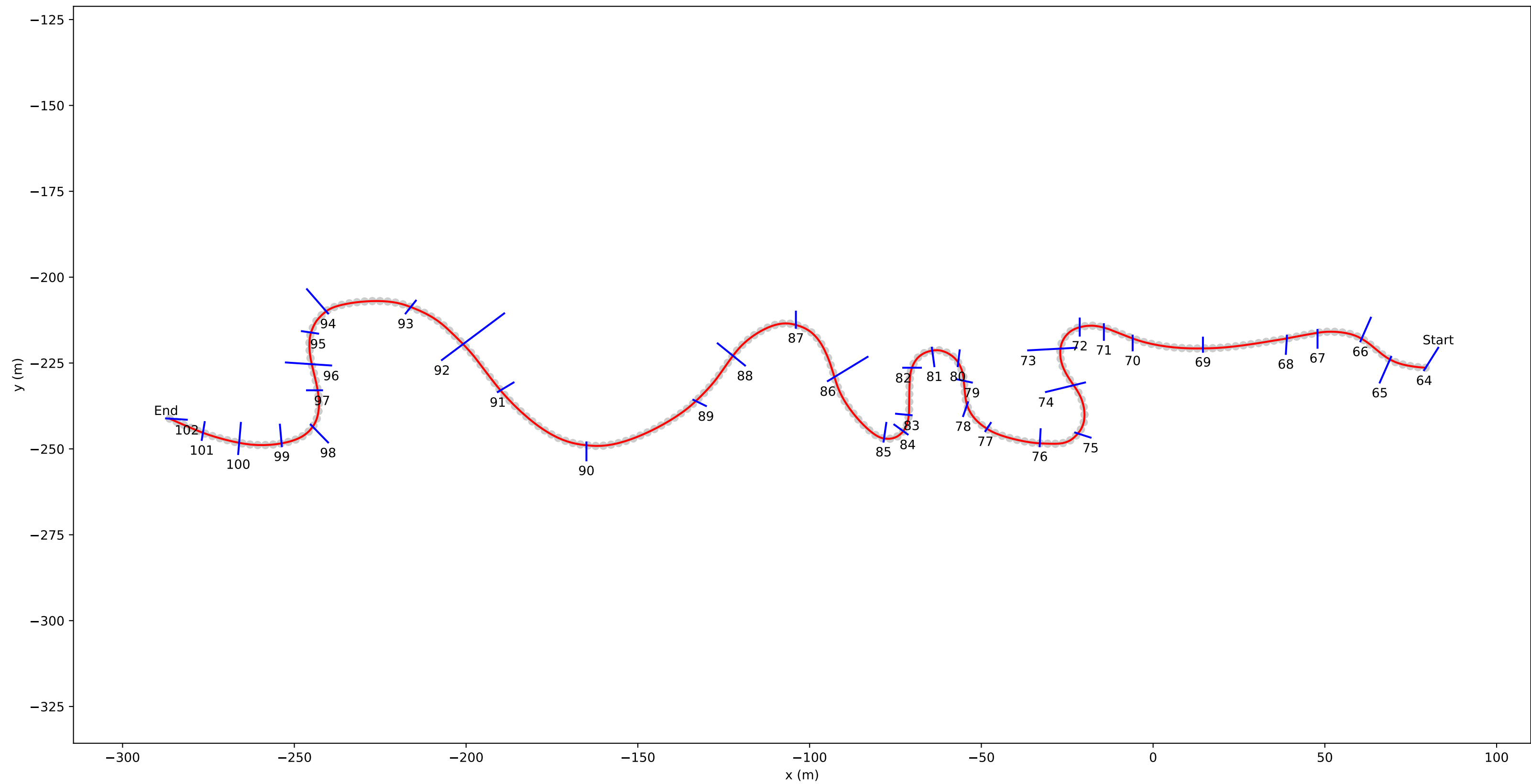


Map plot: optimized path through gates



## Track Generation Report

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Generated: 2026-08-23 03:48:13

Gate file: ref\_less\_gates\_twist.csv

Gate count: 39

Saved DOFs file: opt\_dofs\_2026-08-23\_03-48-13.npy

Saved DOF history file: dof\_history\_2026-08-23\_03-48-13.npy

### Optimization parameters

method: L-BFGS-B

maxiter: 500

ftol: 1e-07

### Point counts

path generation points: 234

### Results

total optimization time: 2475.573 s

total objective function evaluations: 37101

optimizer iterations (accepted steps): 57

total path length: 522.7718 m

**total objective function value: 1.918252**

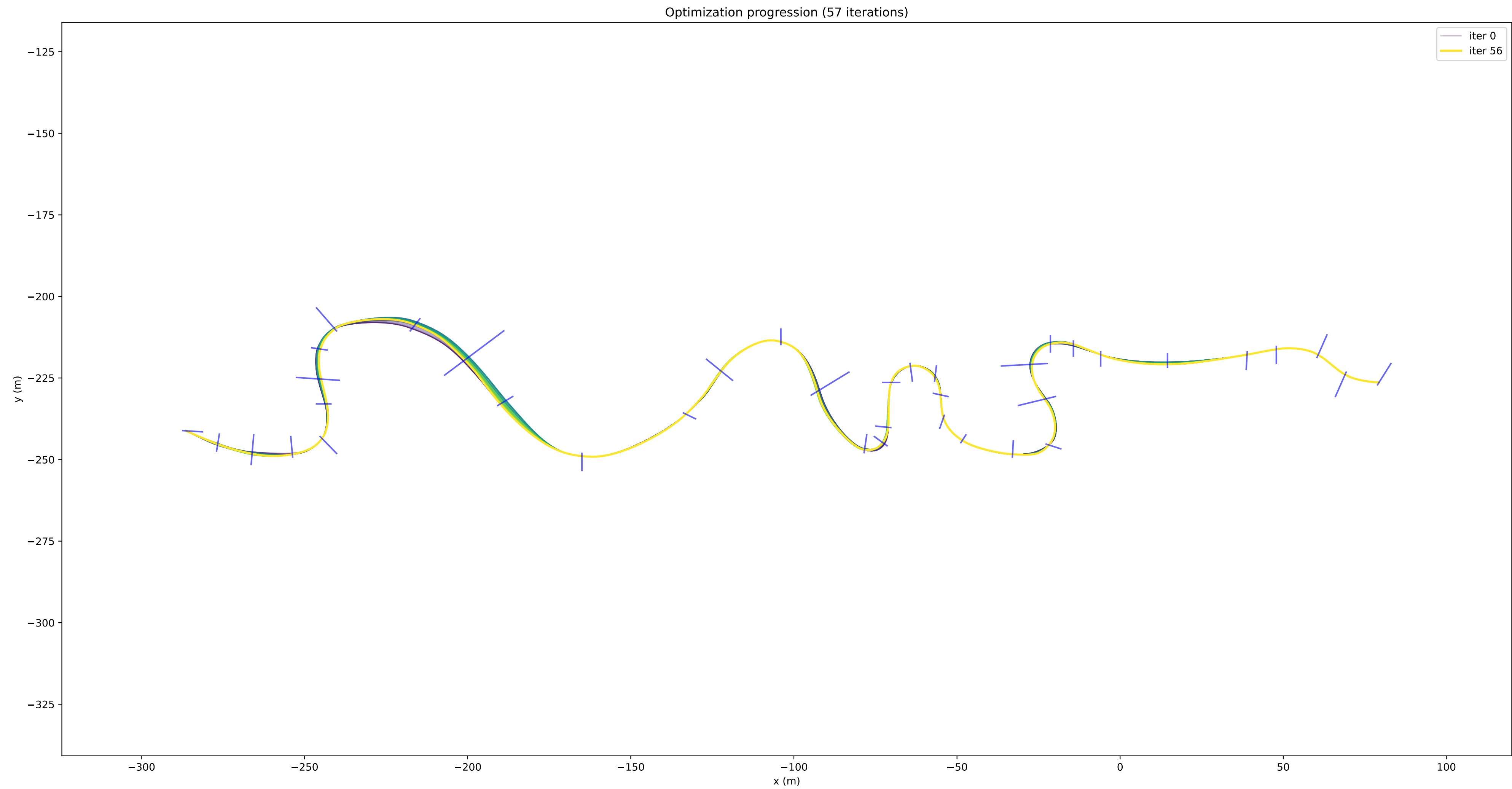
Per-step time estimate (one measured objective evaluation)

make spline from dofs: 6.03 ms

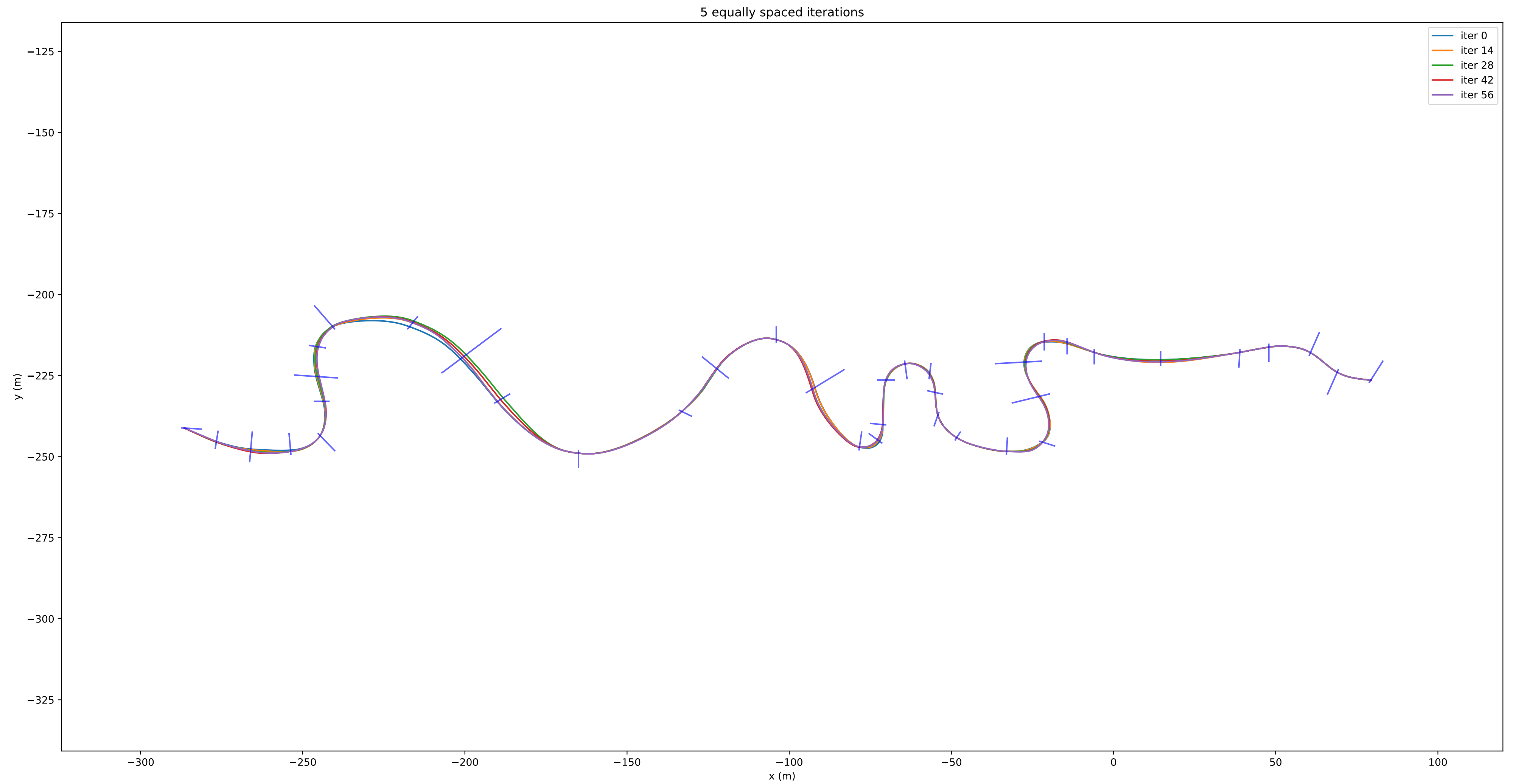
path length function: 0.56 ms

curvature objective function: 73.23 ms

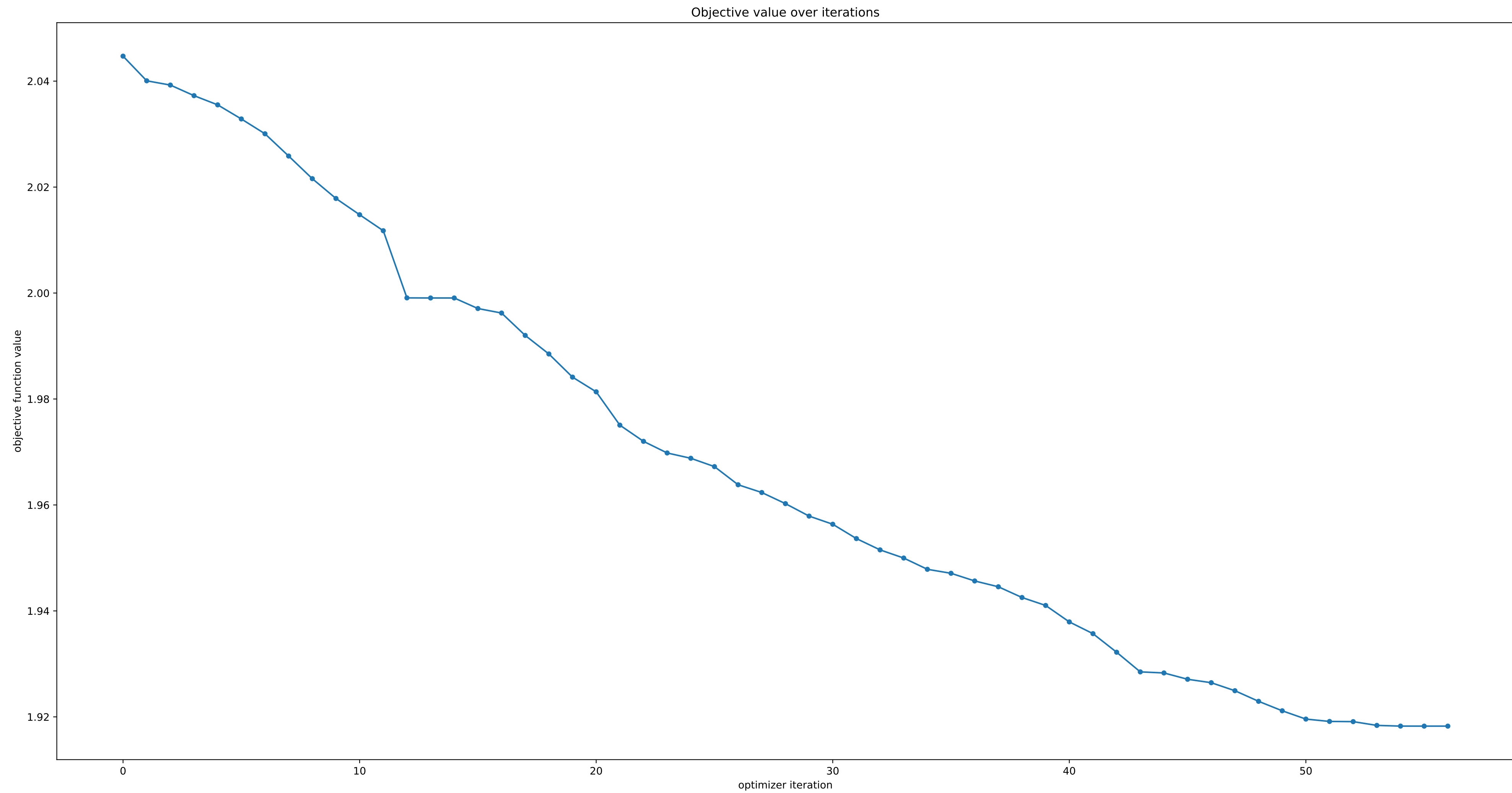
Optimization progression



Equally spaced iterations



Objective value over iterations



Instant radius vs distance

